

SQL (Structured Query Language)

What is Data ?

➤ Data is raw facts and figures (like numbers, text, images) that by itself may not have meaning.

➤ Example: 101, Sagar, 75kg.

Types of data:

1. Structured Data :-

- Data organized in rows and columns (tables),
- Easy to store, search, and manage with SQL
- **Example :**

Student Table:

ID	Name	Age
1	Shubham	22
2	Riya	21

2. Semi-Structured Data :-

- Not in strict rows and columns but has some structure (tags, keys).
- Stored in formats like JSON, XML.
- **Example :**

```
{  
  "Name": "Sagar",  
  "Age" : 22  
}
```

3. Unstructured Data :-

- Raw data with no predefined structure .
- Hard to store in a table; needs special tools.
- **Example:** Text files, images, videos, social media posts,XML.

Database :-

What is a Database ?

- A database is an organized collection of data, stored and managed electronically so it can be easily accessed, managed, and updated,.
- **Example:** A table storing student records

History of SQL :-

1. 1970 – Origin: Dr. E.F. Codd introduced the relational model at IBM.
2. 1974 – Development: IBM began developing SQL (then called SEQUEL) for their System R project.
3. 1979 – Commercial Use: Oracle (then Relational Software Inc.) released the first commercial SQL-based RDBMS.
4. 1986 – Standardization: SQL was adopted as a standard by ANSI (American National Standards Institute).
5. 1987 – ISO Standard: SQL became an international standard (ISO).
6. Ongoing Updates: SQL has evolved with new standards like SQL-92, SQL:1999, SQL:2003, and newer versions.

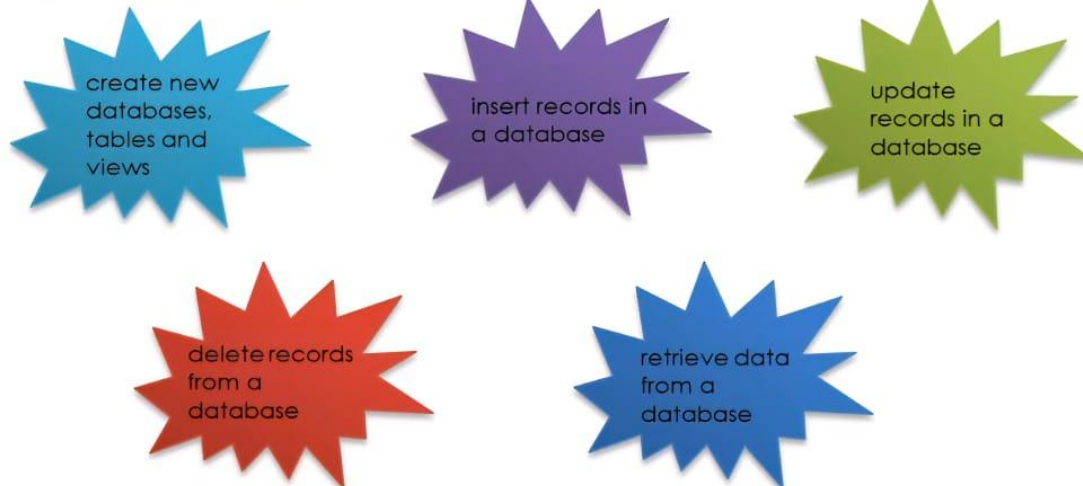
What is SQL ?

- SQL stands for Structured Query Language.
- Designed for managing data in a relational database management system (RDBMS)
- It is pronounced as S-Q-L or sometime See-Qwell.
- SQL is a database language, it is used for database creation, deletion, fetching rows, and modifying rows, etc.
- SQL is based on relational algebra and tuple relational calculus.

Why SQL is Required ?

1. Data Management: SQL helps store, retrieve, and manage data in databases.
2. Standard Language: It is the standard language for relational database systems.
3. Data Manipulation: Allows inserting, updating, and deleting data.
4. Security Control: Supports user access control and permissions.
5. Data Analysis: Helps in analyzing data using powerful queries.

Why SQL is required



SQL Syntax :-

- SQL is not case sensitive.
- SQL keywords are written in uppercase.
- We can place a single SQL statement on one or multiple text lines.
- SQL statements are started with any of the SQL commands/keywords like SELECT, INSERT, UPDATE, DELETE, ALTER, DROP etc. and the statement ends with a semicolon(;).

Important SQL Commands

